A Taxonomy and Probe Design for Hallucinations in LLM-Based Library Assistants

Project Report - Hallucination and Knowledge-Cutoff Probe for a Library Assistant

Abstract - Library assistants built on large language models (LLMs) can answer catalogue, policy and reference questions in natural language, but they can also fabricate book titles, authors, ISBNs, call numbers and citations. This paper proposes a taxonomy of hallucinations specific to the library domain and describes the design of a probe set that elicits and labels them. We define five hallucination types, a three-tier probe construction method (catalogue-grounded, adversarial, and unanswerable queries), and an annotation protocol with inter-annotator agreement. The framework is intended as the foundation for the Hallucination and Knowledge-Cutoff Probe project.

Keywords: hallucination, library assistant, large language models, taxonomy, probe design, catalogue

I. Introduction

Libraries are trusted information intermediaries. When a conversational assistant recommends a book that does not exist or invents a loan rule, the damage goes beyond an ordinary chatbot error: patrons may travel to a branch, request an interlibrary loan, or cite a non-existent source in their coursework. LLMs are known to produce fluent but unfaithful or non-factual text [1], [2], and benchmarks such as TruthfulQA show that larger models can still imitate common falsehoods [3]. However, general benchmarks do not capture the entity-heavy, identifier-heavy nature of bibliographic data.

This paper addresses a narrow question: *how should hallucinations of a library assistant be categorised and probed?* Our contributions are (i) a domain taxonomy, (ii) a probe-construction method tied to a library's own catalogue, and (iii) an annotation protocol that yields reproducible labels.

II. Background

Ji et al. separate *intrinsic* hallucinations (contradicting the source) from *extrinsic* ones (unverifiable from the source) [1]. Huang et al. extend this into input-conflicting, context-conflicting and fact-conflicting categories for LLMs [2]. In a library assistant, the 'source' is usually a catalogue record or a policy page retrieved at query time, so both views are useful: the assistant may contradict a retrieved record (intrinsic) or add facts that no record contains (extrinsic).

Long-tail knowledge is a known weak point. Mallen et al. show that parametric memory fails on rarely seen entities, where retrieval helps most [4]. Most books in a university or public catalogue are long-tail entities, so a library assistant is exactly the setting in which unaided models are weakest.

III. Proposed Taxonomy

We define five hallucination types observed in bibliographic question answering:

Type	Description	Example probe
H1 Phantom item	Invents a book, journal or author that does not exist	'Recommend books by [real author] on topic X'
H2 Metadata corruption	Real item with wrong author, year, edition, ISBN or call number	'Give the ISBN of [real title]'
H3 Citation fabrication	Fake DOI, page range, or reference	'Cite three papers on library classification'
H4 Policy/service invention	Invents loan periods, fines, opening hours, or services	'How many renewals are allowed?'
H5 Availability fabrication	Claims an item is on shelf or available without catalogue evidence	'Is [title] available in the main branch?'

H1-H3 are largely parametric-knowledge errors; H4-H5 are grounding errors, because the correct answer exists only in the library's own live system. This distinction determines which mitigation is appropriate

(Paper 3).

IV. Probe Construction

Tier 1 - Catalogue-grounded probes. Sample records from the library's catalogue (stratified by popularity, language and publication year) and generate questions whose gold answer is the record itself. This yields objective ground truth for H2 and H5.

Tier 2 - Adversarial probes. Combine a real author with a topic the author never wrote on, or ask for 'the second edition' of a title that has only one. A faithful assistant should correct the false premise, which tests H1 and H3.

Tier 3 - Unanswerable probes. Ask for information the library cannot know, such as a non-existent policy or a book not in the collection. The desired behaviour is abstention or redirection to a librarian. This follows the intuition that models can partially estimate what they do not know [5], and tests whether the assistant exposes that uncertainty.

V. Annotation Protocol

Each response is split into atomic claims, in the spirit of FActScore [6]. Two annotators label every claim as *supported*, *contradicted*, or *unverifiable* against the catalogue, and assign a taxonomy type to non-supported claims. Disagreements are resolved by a third annotator. We report Cohen's kappa per type and revise the guidelines until agreement is acceptable. Metrics include claim-level factual precision, response-level hallucination rate per type, and abstention precision on Tier 3.

VI. Threats to Validity and Conclusion

Catalogue records may themselves be wrong, probes may leak into training data, and model behaviour changes across versions, so probe sets must be versioned and re-run. The taxonomy covers English-language bibliographic queries; multilingual collections will need extension. We have presented a taxonomy, probe tiers and an annotation protocol that together let a library measure how often, and in what way, its assistant hallucinates. Paper 2 extends the probes to temporal knowledge, and Paper 3 evaluates mitigations.

References

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